Computation of 2-variable likelihood

Suppose we want to estimate the state variables u and v, given the observations θ and s, where:(following your notation, the order of u and v)

$$\theta = \arctan\left(\frac{v}{u}\right), \quad s = \sqrt{u^2 + v^2}. \tag{1}$$

Let us take 3 mesh grid points as an example to illustrate. We define the state vector $\mathbf{x} \in \mathbb{R}^6$ and observation vector $\mathbf{y} \in \mathbb{R}^6$ as:

$$\mathbf{x} = [u_1, u_2, u_3, v_1, v_2, v_3]^\top, \tag{2}$$

$$\mathbf{y} = [\theta_1, \theta_2, \theta_3, s_1, s_2, s_3]^\top. \tag{3}$$

The observation operator $\mathbf{g}(\mathbf{x}) : \mathbb{R}^6 \to \mathbb{R}^6$ maps the state space to the observation space:

$$\mathbf{y} = \mathbf{g}(\mathbf{x}) + \boldsymbol{\epsilon}, \tag{4}$$

where $\boldsymbol{\epsilon}$ represents the observation error.
Recall our likelihood is defined as: $p(\mathbf{y}|\mathbf{x}) \propto exp[-\frac{1}{2}(\mathbf{g}(\mathbf{x})-\mathbf{y})^T\mathbf{\Sigma}^{-1}(\mathbf{g}(\mathbf{x})-\mathbf{y})]$, then,

$$\nabla \log\ p(\mathbf{y}|\mathbf{x}) = -\mathbf{D}(\mathbf{g}(\mathbf{x}))^T\mathbf{\Sigma}^{-1}(\mathbf{g}(\mathbf{x}) - \mathbf{y}), \tag{5}$$

$$\mathbf{D}(\mathbf{g}(\mathbf{x})) = \frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \begin{bmatrix} -\frac{v_1}{s_1^2} & 0 & 0 & \frac{u_1}{s_1^2} & 0 & 0 \\ 0 & -\frac{v_2}{s_2^2} & 0 & 0 & \frac{u_2}{s_2^2} & 0 \\ 0 & 0 & -\frac{v_3}{s_3^2} & 0 & 0 & \frac{u_3}{s_3^2} \\ \frac{u_1}{s_1} & 0 & 0 & \frac{v_1}{s_1} & 0 & 0 \\ 0 & \frac{u_2}{s_2} & 0 & 0 & \frac{v_2}{s_2} & 0 \\ 0 & 0 & \frac{u_3}{s_3} & 0 & 0 & \frac{v_3}{s_3} \end{bmatrix} \tag{6}$$

,

$$\mathbf{g}(\mathbf{x}) - \mathbf{y} = \begin{bmatrix} \arctan(\frac{v_1}{u_1}) - \theta_1 \\ \arctan(\frac{v_2}{u_2}) - \theta_2 \\ \arctan(\frac{v_3}{u_3}) - \theta_3 \\ \sqrt{v_1^2 + u_1^2} - s_1 \\ \sqrt{v_2^2 + u_2^2} - s_2 \\ \sqrt{v_3^2 + u_3^2} - s_3 \end{bmatrix} \tag{7}$$

thus the size of likelihood for each ensemble should be $(6, 1)$.